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SA toolbox: A Python based data-driven toolbox for global sensitivity analysis

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Summary

SA-Toolbox is a Python-based, data-driven toolkit for performing comprehensive Globa Sensitivity Analysis (GSA) on computational models. It provides a graphical user interface (GUI) built with `tkinter` that enables researchers to explore input-output relationships using a suite of advanced SA methods without requiring programming expertise. The toolbox supports a broad set of techniques for assessing main effects, including regression, variance based and moment independent methods . It also implements High-Dimensional Model Representation (HDMR) for interaction effects and introduces a novel checkerboard partitioning method for estimating third-order interactions. These methods are grounded in established SA frameworks such as `SALib` (Herman & Usher, 2017; Iwanaga et al., 2022) and the `Global Sensitivity Analysis Collection` by Plischke (Plischke, 2022). SA-Toolbox is for applications where only input–output data is available—users simply upload a dataset to begin analysis. This data-driven approach makes it well suited for simulation models such as radioactive waste repository safety assessment, geochemica process , and environmental modeling, where complex simulations are common and model internals may not be directly accessible. By combining methodological breadth with an intuitive interface, SA-Toolbox lowers the barrier to applying rigorous sensitivity analysis in scientific and engineering workflows.

Statement of Need

Sensitivity analysis (SA) is a technique used to study how changes in the inputs of a model affect its outputs. This can help identify which variables have the greatest impact on model behavior. One common form of SA is global sensitivity analysis (GSA), which looks at input-output relationships across the full range of input variability. A widely used approach in GSA is to decompose the output variance into contributions from each input variable (Saltelli et al., 2000).

These methods are implemented in several Python libraries, such as `SALib`. However, most of these libraries require users to write Python code and be familiar with programming concepts such as data structures, loops, and function calls. They also assume knowledge of how each sensitivity method works, including how to prepare inputs, configure parameters, and interpret results. For researchers and practitioners who are not programmers but who work with input–output data from simulations or experiments, these tools may be difficult to use.

SA-Toolbox is a Python-based graphical user interface (GUI) that allows users to perform sensitivity analysis through a point-and-click interface. It is designed to be used without writing code. All steps of the analysis are integrated into the software interface. Users can load data files in tabular format (e.g., CSV), select variables for analysis, choose methods, configure parameters, and view results, all from within the same window.

Once data is loaded, the toolbox allows the user to inspect it using visual tools such as scatter plots, histograms, and correlation matrices. This step can help users understand their data before running sensitivity analysis. The GUI also supports creating subsets of the data or filtering it, which may be useful when working with simulation output from different scenarios.

After preparing the data, users assign roles to variables (e.g., input or output) and select which sensitivity analysis methods to apply. The toolbox includes several first-order methods, including variance-based techniques (e.g., RBD-FAST, Pearson η^2), as well as moment-independent methods (e.g., Borgonovo δ, Copula-based indices, FlavourSi), an overview of these methods can be found in (Swiler et al., 2021, 2025). Users can also estimate interactions between variables using methods such as High-Dimensional Model Representation (HDMR), or apply a checkerboard partitioning approach to explore third-order interactions.

Each method can be configured through dialog windows that expose its parameters, such as the number of bootstraps, harmonics, or partitioning intervals. Tooltips are available to explain the purpose of each method and the meaning of each parameter. Once configured, the analysis can be run with a single button.

The output includes sensitivity index values in table form as well as graphical summaries. These include bar charts of main effects, HDMR heatmaps, copula distance plots, and CUSUNORO visualizations. The toolbox also includes a dedicated visualization for the checkerboard method. Results can be inspected in the GUI and exported in standard formats for documentation or further analysis.

The software workflow—starting from data loading and ending in visualization—is shown in Figure 1.

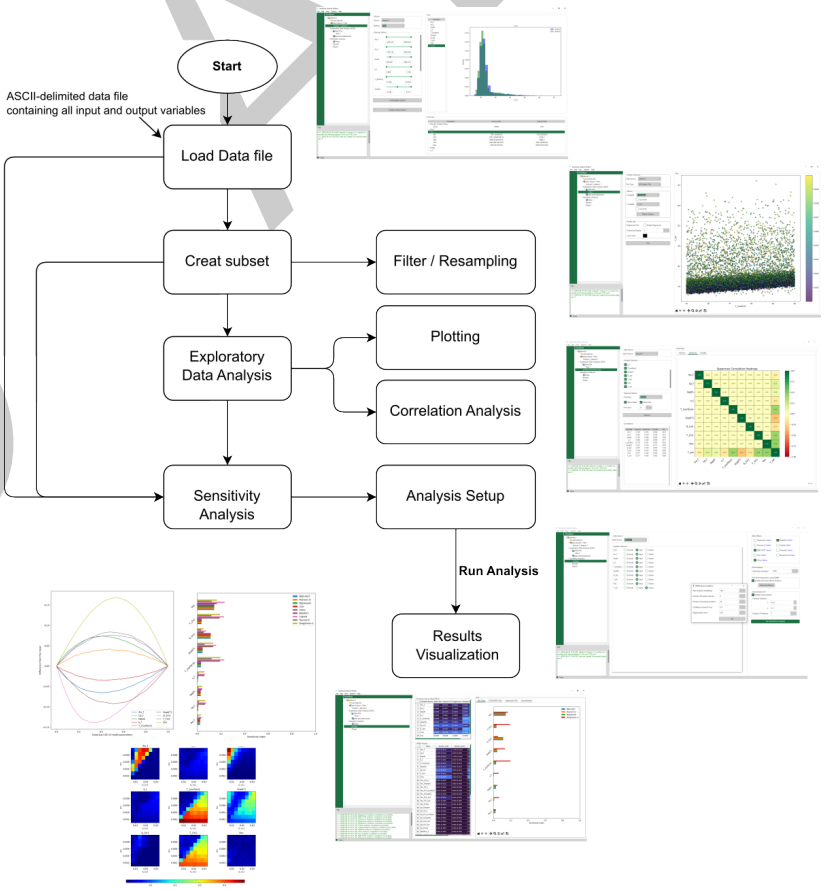

Figure 1: SA-Toolbox workflow from data import to result visualization.

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SA-Toolbox is designed for use cases where a user has a dataset consisting of inputs and corresponding model outputs, but may not have access to the underlying model itself (e.g., black-box simulations, empirical data). It can be used in fields such as environmental modeling, safety assessment, and geochemistry, where large simulation datasets are common (Swiler et al., 2021, 2025). The toolbox does not aim to replace existing code-based sensitivity libraries. Instead, it provides an alternative interface for users who prefer to work visually or who want to conduct sensitivity analysis without programming.

By organizing the full analysis process into a step-by-step graphical workflow and supporting multiple sensitivity methods with adjustable settings, SA-Toolbox can make sensitivity analysis more accessible to a wider range of users. This includes researchers, engineers, and analysts who may not be familiar with Python programming but still need to interpret the influence of model inputs on outputs.

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